<u>1A.</u>

Basis:

i = 1 and 4^1 = 4, therefore P(1) is true and the base case holds since 4 ∈ S

Inductive Step:

Assume that k ∈ Z+ and 4^k ∈ S, prove that 4^k+1 ∈ S.

4^(k+1) = 4^k x 4^1

By IH, 4^k ∈ S and from the base case and given, 4 ∈ S.

From the other given where if s ∈ S and t ∈ S, then st ∈ S, it can be applied to this problem to observe that since 4^k ∈ S and 4 ∈ S, 4^k + 1 = 4^k x 4 ∈ S.

Through MI, it has been shown that for every i ∈ Z+, 4^i ∈ S. So, A ⊆ S.

<u>1B.</u>

Basis:

By basis step of ind. def of S, 4 ∈ S. Prove that S ∈ 4. Since 4 = 4 * 1 and 1 ∈ Z+, by def. of A, 4 ∈ S.

Inductive Step:

Consider x, y ∈ S. Assume x, y ∈ A.

By the inductive step of inductive def, st ∈ S. Prove that st ∈ A.

Since by IH, s, t ∈ A, follows that s = 4a and t = 4b, where s, t, st ∈ Z+.

st = (4(s))(4(t)) → 16(st) → 4(4st)

4(4st) ∈ A because 4 * 4 = 16 which is a multiple of 4, also belongs to A when i = 2.

st ∈ A and st ∈ S, therefore S ⊆ A through strong induction based on the basis and inductive steps above.

<u>2.</u>

Basis:

ε (Empty string) is a palindrome on its own.

a, b, c are all palindromes on their own.

Inductive Step:

If x is a palindrome representing a letter or multiple letters in the alphabet, then any combination of letters like axa, bxb, or cxc are all palindromes. This represents how for any character, if you append x (a character in the alphabet) to the beginning and end of an empty string or palindrome, then it will result in still being a palindrome.

Therefore by strong induction, it is proven that if you add a character that belongs to the alphabet to the beginning and end of a palindrome, or in an exact order from the middle, then by IH, the resulting string will still be a palindrome.

3.
Basis:
Smallest value of k and m in Z+ are 1, so base case must be $2^1 \times 3^1 = 6$.
Therefore, 6 belongs to S.
Inductive Step:
Assume that there exists a number c where $c = 2^k 3^m$ and belongs to S.
Substitute k+1 and/or m+1 into the equation to construct the set even more.
If $c \in S$, then $2c = 2^{k+1} \times 3^m$ is also in S since $2^{(k+1)} = 2 \times 2^k$. Therefore multiples of 2 also belong to the set.
If $c \in S$, then $3c = 2^k \times 3^{m+1}$ is also in S since $3^{(k+1)} = 3 \times 3^k$. Therefore multiples of 3 also belong to the set.
Therefore, set can be reached by using inductive steps to access different elements of S, mainly the multiples of 2, 3, and therefore 6 (and beyond using a combination of multiples).

4.
N(t) = number of vertices in tree T
L(t) = number of leaves in tree T
T is FBT.
Prove for all FBTs T, $n(T) = 2l(T) - 1$
Basis:
FBT with just the root node.
$N(1) = 1$ and $L(1) = 1$ because the root itself is both a node and technically a leaf of the FBT.
Inductive Step:
Prove that if $l + 1$, then N will equal $2(l + 1) - 1$).
Let T have $l + 1$ leaves.
If a leaf l at the bottom of the tree (highest depth) was removed, along with its counterpart leaf, then the parent (vertice) node of that leaf will become a leaf itself.
Therefore, this tree must have $2L - 1$ total nodes
If l and its counterpart leaf are added back to the tree as siblings of the parent node, the parent node will no longer be a leaf and instead will just be a node.
Therefore, the tree loses a leaf but also gains two more leaves in the process, so total leaves is equal to $l + 1$.
Therefore, we have $2l - 1 + 2$, equaling to $2l + 1$, and equaling to $2(l + 1) - 1$
Finally, through induction I have proven that a FBT with l leaves and n vertices will have $N = 2L - 1$ vertices/nodes.

5A.

Basis:

Robot starts at (0, 0).

Belongs in L because (0-0) % 4 = 0, so origin (0, 0) is in L’.

Inductive Step:

Assume robot is at point (x, y) and wants to move to new point

Must increment/decrement either x or y by 4 in order to maintain the mod 4 requirement, so…

Case #1:

x’ = x + 4, y’ = y

(x-y) after the transformation will still be a multiple of 4.

Case #2:

x’ = x and y’ = y + 4

(x-y) after the transformation will still be a multiple of 4.

Case #3:

x’ = x - 4 and y’ = y

(x-y) after the transformation will still be a multiple of 4.

Case #4:

x’ = x and y’ = y - 4

(x-y) after the transformation will still be a multiple of 4.

Therefore, L’ = {(4x, 4y) | c ∈ Z} as the robot will move in multiples of 4 in order to maintain the requirement by L. (0, 0) also checks out too.

5B.

Basis:

Point (0, 0) is in L’ and L because the robot starts there (4(0), 4(0)) = (0, 0) and 0 mod 4 equals 4.

Inductive Step:

Prove that if (x, y) belongs to L’, then (x + 4, y), (x, y + 4), (x - 4, y) and (x, y - 4) are also in L.

As proven previously, every alteration to the coordinates still result in a coordinate pair that belongs in L.

Since every point of L’ is included in L, L’ is a subset of L. So L’ has been proven inductively that it is a subset of L.

5C.